WORKSHEET

Moving medians

1 For each of the following tables of time series data:

i complete the table using three-point median smoothing

ii use CAS or a spreadsheet to plot the original and the smoothed data on the same axes.

a Ticket sales to the drama night at Hopston High School

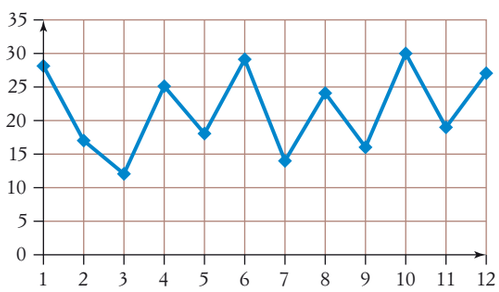
Year	Sales	Three-point moving median
2005	456	
2006	395	
2007	513	
2008	555	
2009	516	
2010	578	
2011	603	

b Wallaby population at Stoney Creek Reserve

Year	Population	Three-point moving median
1980	461	
1985	378	
1990	270	
1995	414	
2000	453	
2005	516	
2010	474	

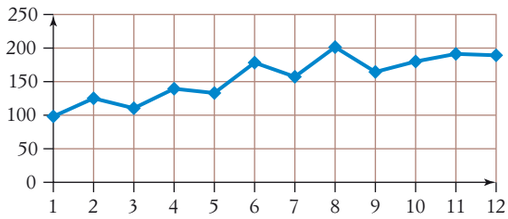
2 For each graph below, draw a smoothed time series plot using the graphical approach to three-point median smoothing.

a



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b



3 For each of the following tables of data:

i complete the table by calculating the two- and four-point moving average with centring to smooth the data

ii graph the original and the smoothed data in a time series plot.

a Production of cholesterol-lowering tablets

Year	No. of tablets (1000s)	Four-point moving median	Centred
2007	140		
2008	250		
2009	275		
2010	245		
2011	280		
2012	325		
2013	310		
2014	320		

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b Average age of the members of a bowling club from 2002 to 2012

Year	Average age	Four-point moving median	Centred
2002	71		
2003	72		
2004	63		
2005	68		
2006	61		
2007	52		
2008	62		
2009	57		
2010	54		
2011	57		
2012	64		

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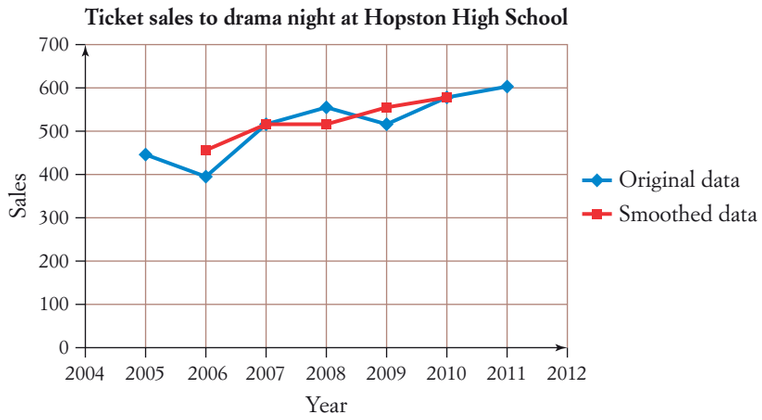
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Answers

1 a i

Year	Sales	Three-point moving median
2005	456	
2006	395	Median of 456, 395, 513 = 456
2007	513	Median of 395, 513, 555 = 513
2008	555	Median of 513, 555, 516 = 516
2009	516	Median of 555, 516, 578 = 555
2010	578	Median of 516, 578, 603 = 578
2011	603	

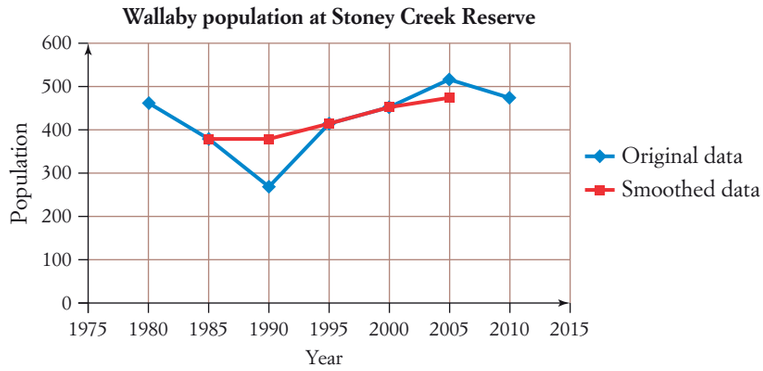
ii



b i

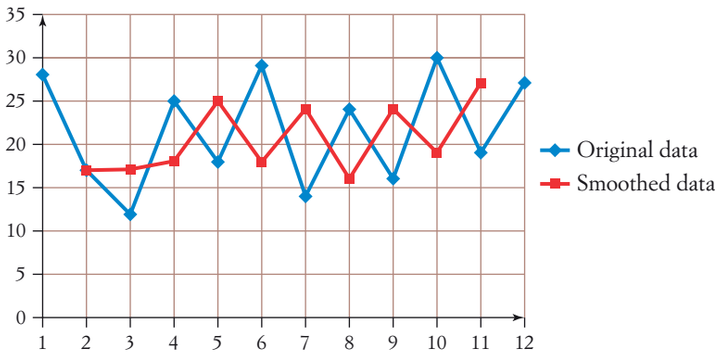
Year	Population	Three-point moving median
1980	461	
1985	378	Median of 461, 378, 270 = 378
1990	270	Median of 378, 270, 414 = 378
1995	414	Median of 270, 414, 453 = 414
2000	453	Median of 414, 453, 516 = 453
2005	516	Median of 453, 516, 474 = 474
2010	474	

ii

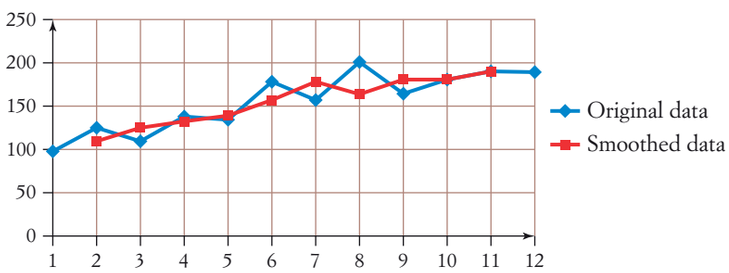


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2 a



b



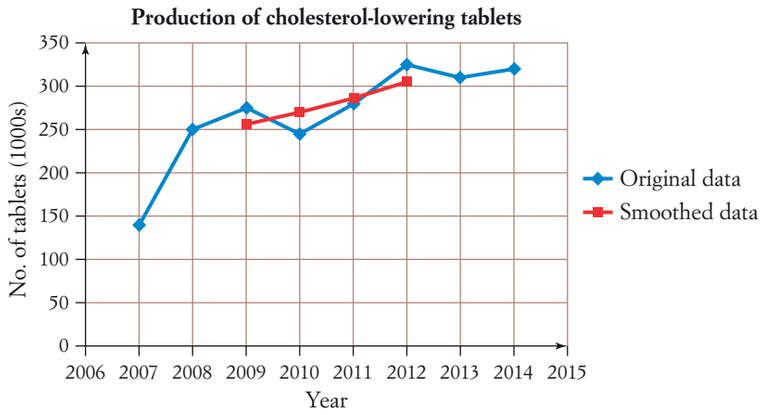
3 a i

Year	No. of tablets (1000s)	Four-point moving median	Centred
2007	140		
2008	250		
		Median of 140, 250, 275, 245 = 247.5	
2009	275		(247.5 + 262.5) ÷ 2 = 255
		Median of 250, 275, 245, 280 = 262.5	
2010	245		(262.5 + 277.5) ÷ 2 = 270
		Median of 275, 245, 280, 325 = 277.5	
2011	280		(277.5 + 295) ÷ 2 = 286.25
		Median of 245, 280, 325, 310 = 295	
2012	325		(295 + 315) ÷ 2 = 305
		Median of 280, 325, 310, 320 = 315	
2013	310		
2014	320		

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ii



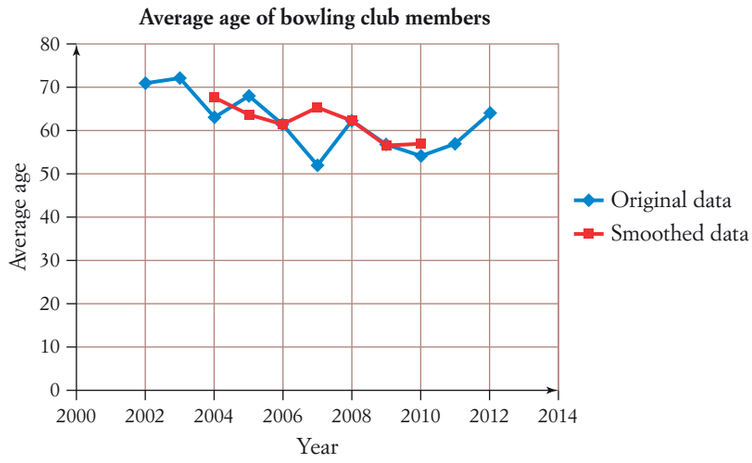
b i

Year	Average age	Four-point moving median	Centred
2002	71		
2003	72		
		Median of 71, 72, 63, 68 = 69.5	
2004	63		(69.5 + 65.5) ÷ 2 = 67.5
		Median of 72, 63, 68, 61 = 65.5	
2005	68		(65.5 + 62) ÷ 2 = 63.75
		Median of 63, 68, 61, 52 = 62	
2006	61		(62 + 61.5) ÷ 2 = 61.75
		Median of 68, 61, 52, 62 = 61.5	
2007	52		(61.5 + 58) ÷ 2 = 59.75
		Median of 61, 52, 62, 57 = 58	
2008	62		(58 + 55.5) ÷ 2 = 56.75
		Median of 52, 62, 57, 54 = 55.5	
2009	57		(55.5 + 57) ÷ 2 = 56.25
		Median of 62, 57, 54, 57 = 57	
2010	54		(57 + 57) ÷ 2 = 57
		Median of 57, 54, 57, 64 = 57	
2011	57		
2012	64		

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ii



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